4-4 Counting Rules

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Everything so far needed `n(E)` and `n(S)`. When the sample space is too large to list, you *count* it instead. The four main counting rules are: the **Fundamental Counting Rule**, **Factorial Notation**, **Permutations**, and **Combinations**. This lesson and §4-5 together carry In-Class Quiz II, and §4-5 is where every rule below gets used.

1 · The Fundamental Counting Rule

FUNDAMENTAL COUNTING RULE

In a sequence of n events in which the first has k_1 possibilities, the

$k_1 \times k_2 \times \cdots \times k_n$

الشرح بالعربي — قاعدة العدّ الأساسية

إذا كان لديك سلسلة من الاختيارات — الأول له k_1 طريقة، والثاني k_2، والثالث k_3 ... — فعدد كل الاحتمالات = **حاصل ضربها**.

مثال: 7 ألوان × 2 نوع × 3 ملمس × 2 استخدام = 84 نوع دهان.

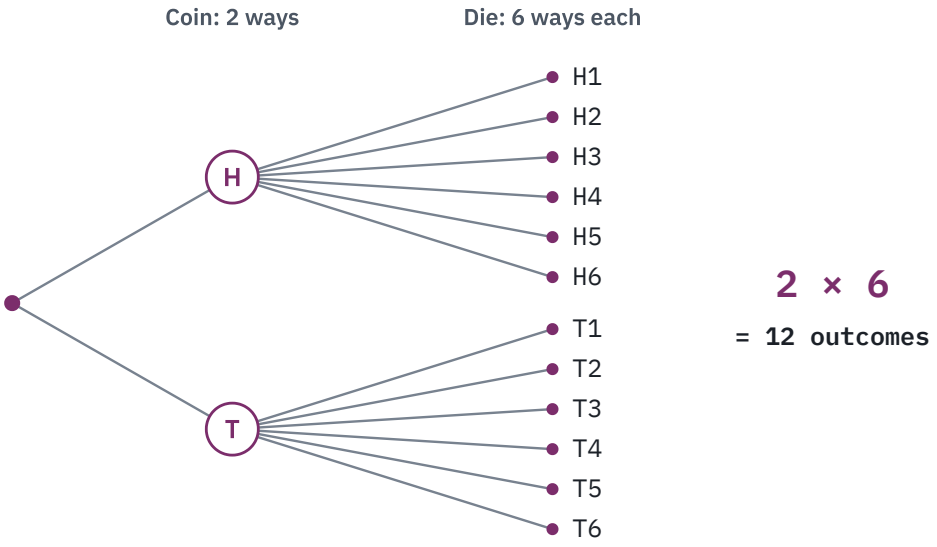
- **مع التكرار** (**repetition allowed**): الأرقام تبقى كما هي من اليسار لليمين ← $10 \times 10 \times 10 \times 10 = 10{,}000$.
- **بدون تكرار** (**not allowed**): الأرقام تنقص واحدًا في كل خانة ← $10 \times 9 \times 8 \times 7 = 5{,}040$.

EXAMPLE 4-38 · TOSSING A COIN AND ROLLING A DIE

A coin is tossed and a die is rolled. Find the number of outcomes for the sequence of events.

$2 \times 6 =$ **12 outcomes**

Namely H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6.



The Fundamental Counting Rule as a tree: each of the 2 coin results branches into the same 6 die results, so the total is 2 × 6 = 12 — multiply, don't add.

قاعدة العدّ الأساسية كشجرة: كل نتيجة من نتيجتي العملة تتفرع إلى 6 نتائج للنرد، فالمجموع 2 × 6 = 12 — نضرب ولا نجمع.

EXAMPLE 4-39 · TYPES OF PAINT

A paint manufacturer wishes to manufacture several different paints. The categories include — Color: red, blue, white, black, green, brown, yellow. Type: latex, oil. Texture: flat, semigloss, high gloss. Use: outdoor, indoor. How many different kinds of paint can be made if you select one color, one type, one texture, and one use?

Count the options in each category: 7 colors, 2 types, 3 textures, 2 uses.

$7 \times 2 \times 3 \times 2 =$ **84 different kinds of paint**

EXAMPLE 4-40 · DISTRIBUTION OF BLOOD TYPES

There are four blood types: A, B, AB, O. Blood can also be Rh+ or Rh−. A blood donor can be classified as either male or female. How many different ways can a donor have his or her blood labeled?

$4 \times 2 \times 2 =$ **16 ways**

EXAMPLE 4-41 · RAILROAD MEMORIAL LICENSE PLATES

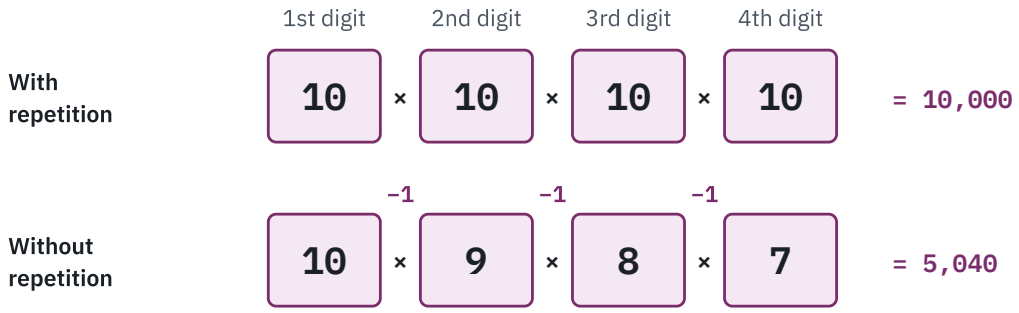
A license plate has a four-digit identification number. How many different plates can be made if: (a) digits are from 0 to 9 and repetitions are permitted? (b) digits are from 0 to 9 and repetitions are **not** permitted?

1. **(a) With repetition.** Every slot still has all 10 digits available:

   $10 \times 10 \times 10 \times 10 = 10^4 =$ **10,000 plates**

2. **(b) Without repetition.** Each digit used is removed from the pool:

   $10 \times 9 \times 8 \times 7 =$ **5,040 plates**



One box per digit, with the number of choices written inside. With repetition every slot keeps all 10 digits. Without repetition each slot has one fewer choice than the slot before.

خانة لكل رقم وبداخلها عدد الخيارات. مع التكرار: كل خانة 10 خيارات. بدون تكرار: كل خانة أقل بواحد من التي قبلها 10)، 9، 8، (7.

THE LECTURER'S OWN RULE OF THUMB

"If repetitions are permitted, then the numbers **stay the same** going from left to right. If repetitions are not permitted, then the numbers **decrease by 1** for each place left to right." Write this on your formula sheet exactly as stated — it answers half of all counting-rule questions on sight.

2 · Factorial notation

FACTORIAL

$$n! = n \times (n-1) \times (n-2) \times \cdots \times (2 \times 1)$$

$$0! = 1$$

Examples: $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ · $9! = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 362{,}880$

الشرح بالعربي — المضروب

المضروب `n!` (يُقرأ **n factorial**) هو حاصل ضرب كل الأعداد من `n` نزولًا إلى `1`، مثل `5! = 5×4×3×2×1 = 120`.

وهو يساوي **عدد طرق ترتيب n أشياء مختلفة في صف**.

تعريف مهم جدًا: `0! = 1` وليس صفرًا.

FACTORIALS WORTH KNOWING COLD

N	0	1	2	3	4	5	6	7	8	9	10
n!	1	1	2	6	24	120	720	5,040	40,320	362,880	3,628,800

Casio keystrokes. To compute `(5 + 3)! = 40,320`: `( 5 + 3 )  SHIFT  x⁻¹ (x!)  =`

TRAP — 0! = 1, NOT 0

This matters because ${}_nP_n = n!/(n-n)! = n!/0! = n!$ and ${}_nC_n = n!/(0! \cdot n!) = 1$. If you set 0! = 0 you get a division by zero and lose the whole question. Memorise it as a definition, not as a calculation.

3 · Permutations — order matters

A **permutation** is an arrangement of *n* objects in a **specific order**.

PERMUTATION RULE 1 — N OBJECTS TAKEN R AT A TIME

$${}_nP_r = \frac{n!}{(n-r)!}$$

Casio: to compute ${}_7P_4$ → `7  SHIFT  × (nPr)  4  =`

الشرح بالعربي — التباديل (الترتيب مهم)

`nPr` = عدد طرق اختيار `r` من `n` **مع ترتيبها**. نستخدمها عندما يكون **الترتيب مهمًا**.

علامات أن الترتيب مهم: ترتيب أو تصنيف (**rank**)، مراكز (أول/ثاني/ثالث)، مناصب مختلفة (رئيس/نائب/أمين سر)، ترتيب الظهور، خانات مسمّاة (خريف/ربيع)، أرقام كود أو لوحة.

مثال: اختيار 3 ضيوف من 6 مع أهمية ترتيب ظهورهم = `6P3 = 6!/3! = 120` .

EXAMPLE 4-43 · BUSINESS LOCATION

(a) A business owner has a choice of 5 locations in which to establish her business. She decides to rank each location according to certain criteria, such as price of the store and parking facilities. How many different ways can she rank the 5 locations?
(b) Suppose she wishes to rank only the **top 3 locations** selected from 5 locations. How many different ways can she rank them?

1. **(a)** All 5 ranked, order matters: $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 =$ **120 ways**

   Equivalently: ${}_5P_5 = 5!/0! = 120/1 = 120$

2. **(b)** 3 chosen from 5, order matters: ${}_5P_3 = \dfrac{5!}{(5-3)!} = \dfrac{120}{2} =$ **60 ways**

EXAMPLE 4-44 · RADIO SHOW GUESTS

A radio talk show host can select 3 of 6 special guests for her program. The **order of appearance** of the guests is important. How many different ways can this be done?

"Order of appearance is important" → permutation.

$${}_6P_3 = \frac{6!}{(6-3)!} = \frac{720}{6} =$$ **120 ways**

EXAMPLE 4-45 · SCHOOL MUSICAL PLAYS

A school musical director can select 2 musical plays to present next year. One will be presented in the fall, and one will be presented in the spring. If she has 9 to pick from, how many different possibilities are there?

The two slots are *distinguishable* (fall vs spring), so order matters → permutation.

$$ {}_9P_2 = \frac{9!}{(9-2)!} = \frac{362{,}880}{5{,}040} = \textbf{72 possibilities} $$

Why not a combination? Play X in fall + play Y in spring is a different programme from Y in fall + X in spring. Labelled slots always mean permutation.

EXAMPLE 4-42 · LOCK CODES

A person needs to make up a four-digit combination to open the lock on the garage door. The person decides to use four **different** digits from the first four digits, 0 through 3. How many different combinations can be made?

Four distinct digits arranged in all four positions — a full permutation:

$$ 4! = 4 \cdot 3 \cdot 2 \cdot 1 = \textbf{24 combinations} $$

Note the vocabulary trap: everyday English calls this a "combination lock", but mathematically it is a *permutation* — 0123 and 3210 open different locks. The word "combination" in a question is never evidence; only the question "does order matter?" decides.

PERMUTATION RULE 2 — WHEN SOME OBJECTS ARE IDENTICAL

$$ {}_nP_{r_1\ r_2\ r_3\ \ldots\ r_p} = \frac{n!}{r_1! \cdot r_2! \cdot r_3! \cdots r_p!} $$

where $r_1 + r_2 + \cdots + r_p = n$. Divide out the arrangements you cannot tell apart.

الشرح بالعربي — تباديل مع عناصر متكررة

إذا كانت بعض الأشياء **متطابقة** (مثل الحروف المكررة في كلمة)، نقسم `n!` على مضروب عدد تكرار كل عنصر، لأن تبديل عنصرين متطابقين لا يعطي ترتيبًا جديدًا.

مثال: كلمة STATISTICS فيها 10 حروف: S ثلاث مرات، T ثلاث مرات، I مرتان ← `10! ÷ (3! × 3! × 2!) = 50,400`.

نصيحة: عدّ الحروف أولًا وتأكد أن مجموع التكرارات يساوي `n`.

EXAMPLE 4-46 · LETTER PERMUTATIONS

How many permutations of the letters can be made from the word *STATISTICS*?

1. **Count the letters:** S-T-A-T-I-S-T-I-C-S → n = 10.
2. **Tally repeats:** S appears 3 times, T appears 3 times, I appears 2 times, A once, C once. Check: 3 + 3 + 2 + 1 + 1 = 10 ✔
3. $$\text{Permutations} = \frac{10!}{3! \cdot 3! \cdot 2! \cdot 1! \cdot 1!} = \frac{3{,}628{,}800}{6 \cdot 6 \cdot 2} = \frac{3{,}628{,}800}{72} =$$

HW 30 · TOOTHPASTE DISPLAY

How many different ways can 5 identical tubes of tartar control toothpaste, 3 identical tubes of bright white toothpaste, and 4 identical tubes of mint toothpaste be arranged in a grocery store counter display?

1. n = 5 + 3 + 4 = 12 tubes; three identical groups of sizes 5, 3, 4.
2. $$= \frac{12!}{5! \cdot 3! \cdot 4!} = \frac{479{,}001{,}600}{120 \cdot 6 \cdot 24} = \frac{479{,}001{,}600}{17{,}280} = \boxed{27{,}720 \text{ ways}}$$

4 · Combinations — order does not matter

A selection of distinct objects **without regard to order** is called a **combination**.

COMBINATION RULE

$$ {}_nC_r = \frac{n!}{(n-r)! \cdot r!} $$

Important: in a combination, the order or arrangement of the objects is *not* important; order *is* important in a permutation. Casio: to compute ${}_{10}C_4$ → `10 SHIFT ÷ (nCr) 4 =`

الشرح بالعربي — التوافيق (الترتيب غير مهم)

`nCr` = عدد طرق اختيار `r` من `n` **بدون ترتيب** — مجرد اختيار.

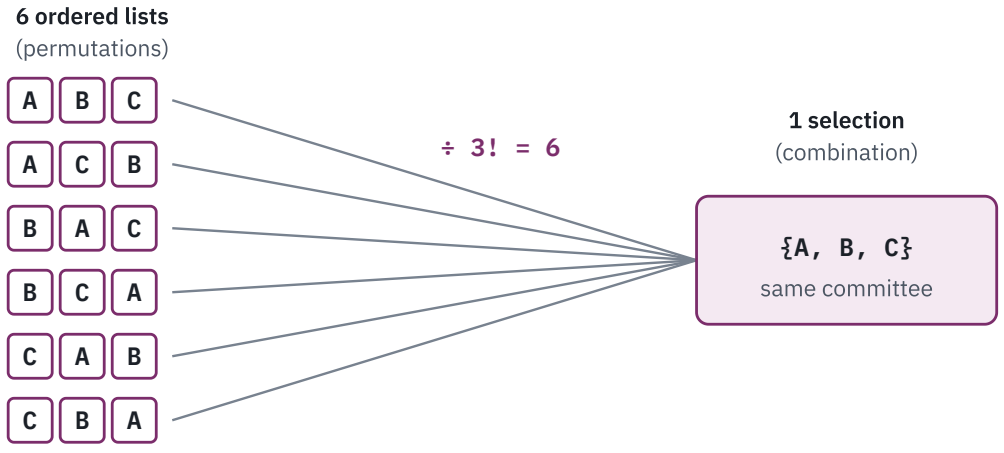
علامات أن الترتيب غير مهم: لجنة بدون مناصب، فريق، مجموعة، اختيار كتب أو أفلام أو مواضيع، سحب عدة قطع من صندوق دفعة واحدة.

العلاقة: `nCr = nPr ÷ r!` ، لذلك التوافيق دائمًا أصغر من أو تساوي التباديل.

مثال: اختيار 4 أفلام من 10 = `10C4 = 210` . وإذا كان الاختيار من مجموعتين (3 نساء من 7 **و** رجلان من 5) فإننا **نضرب**: `7C3 × 5C2 = 35 × 10 = 350` .

THE RELATIONSHIP WORTH SEEING ONCE

${}_nC_r = {}_nP_r \div r!$ — a combination is a permutation with the `r!` internal orderings divided out. That is why ${}_nC_r$ is always ≤ ${}_nP_r$. If your combination answer comes out *larger* than the matching permutation, you have used the wrong formula.



The same three people in all 3! = 6 orders. As a ranked list they are 6 different results (permutations); as a committee they are one result (combination). That is why nCr = nPr ÷ r!.

نفس الأشخاص الثلاثة بـ 6 ترتيبات مختلفة. كقائمة مرتّبة هي 6 نتائج (تباديل)، وكلجنة هي نتيجة واحدة (توافيق). لذلك nCr = nPr ÷ r!.

EXAMPLE 4-48 · MOVIES AT THE PARK

The director of Movies at the Park must select 4 movies from a total of 10 movies to show on Movie Night at the Park. How many different ways can the selections be made?

Just a selection — nothing says the four are shown in a particular order → combination.

$$ {}_{10}C_4 = \frac{10!}{(10-4)! \cdot 4!} = \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} = \frac{5{,}040}{24} = \boxed{210 \text{ ways}} $$

EXAMPLE 4-49 · COMMITTEE SELECTION

In a club there are 7 women and 5 men. A committee of 3 women and 2 men is to be chosen. How many different possibilities are there?

1. **Choose the women:** ${}_7C_3$ = 7!/(4!·3!) = (7·6·5)/(3·2·1) = 210/6 = **35**
2. **Choose the men:** ${}_5C_2$ = 5!/(3!·2!) = (5·4)/2 = **10**
3. **Combine with the Fundamental Counting Rule** (a choice of women *and* a choice of men):

   $35 \times 10 =$ **350 possibilities**

TRAP — "AND" MEANS MULTIPLY, NOT ADD

Choosing 3 women **and** 2 men = $35 \times 10 = 350$. Students who write 35 + 10 = 45 are answering a different question ("choose 3 women *or* 2 men"). Within counting rules, **and → ×, or → +.**

5 · Summary of the counting rules (Table 4-1)

TABLE 4-1 — SUMMARY OF COUNTING RULES

RULE	DEFINITION	FORMULA
Fundamental counting rule	The number of ways a sequence of n events can occur if the first event can occur in k_1 ways, the second in k_2 ways, etc.	$k_1 \cdot k_2 \cdot k_3 \cdots k_n$
Permutation rule 1	The number of permutations of n objects taking r objects at a time (**order is important**).	${}_nP_r = n! / (n-r)!$

RULE	DEFINITION	FORMULA
Permutation rule 2	The number of permutations of n objects when r_1 objects are identical, r_2 objects are identical, ..., r_p objects are identical.	$n! / (r_1!\ r_2!\ \cdots\ r_p!)$
Combination rule	The number of combinations of r objects taken from n objects (**order is not important**).	${}_nC_r = n! / [(n - r)!\ r!]$

القاعدة الحاسمة: Permutation أم Combination؟

اسأل: **هل يتغير الناتج إذا غيّرت الترتيب؟**

- **نعم** (ترتيب، تصنيف، مراكز أول/ثاني، مناصب مختلفة، أرقام لوحة) ← **Permutation** `nPr`
- **لا** (لجنة، مجموعة، اختيار عدد من أشياء بدون أدوار) ← **Combination** `nCr`

وتذكّر: كلمة "**combination lock**" في اللغة الدارجة هي في الحقيقة **permutation** رياضيًا، فلا تعتمد على الكلمة — اعتمد على السؤال.

Chapter 4 study series — 11 files. Read the lessons in order; the revision files at the end draw on all of them.

00 · Exam brief — what the outline tests

01 · Lesson 4-1 — Sample Spaces and Probability

02 · Lesson 4-2 — The Addition Rules

03 · Lesson 4-3 — Multiplication Rules and Conditional Probability

04 · Lesson 4-4 — Counting Rules ← this file

05 · Lesson 4-5 — Probability and Counting Rules

06 · Master formula sheet

07 · Which rule do I use?

08 · The 14 traps that cost marks

09 · Review exercises, solved

10 · The A+ study protocol

Rebuilt from Dr. Ahmed Al-Hadhrami's lecture slides (Spring 24) and aligned to the SQU College of Science STAT 2101 course outline, Fall 2026. Textbook: Allan G. Bluman, *Elementary Statistics: A Step by Step Approach*, brief version, 10th edition, McGraw Hill (2017). A companion 100-question bank with full solutions accompanies this series.